How Colors and Gases Could Help Us Find Tiny Aliens

From: Science News Explores Mined: 2026-10-12 3 min read

If there is life on other worlds, it is probably very tiny. Scientists think it may be **microbes**, which are living things with just one cell.

Venus is very hot on the ground. But high in its clouds, it is more like Earth. Scientists have seen signs of some gases there. Some think **microbes** could live in those clouds.

Scientist Noelle Bryan found **microbes** living very high above Earth. Some of them make bright colors that protect them from strong sunlight. Another scientist, Lígia Coelho, studied those colors. Future **telescopes** might look for the same colors on faraway planets.

Scientists are also thinking about Mars. Andrew Schuerger and Rachel Harris found a microbe on Earth that can make **methane** gas even when the air is thin like on Mars. **Rovers** have found **methane** on Mars. Schuerger says a **rover** might only need to dig about 8 inches to find Martian **microbes**, if they exist.

Word Treasure

microbes -- Living things so tiny you need a microscope to see them.
telescopes -- Tools that help us see things very far away.
methane -- A kind of gas that some tiny living things make.
rover -- A robot vehicle that drives on another planet.

Background you need

Scientists are searching for life on other planets and moons.

The first alien life we find might be tiny microbes, not big creatures.

Rovers are robots that explore Mars and send information back to Earth.

Break it down

WHO: Scientists Noelle Bryan, Lígia Coelho, Andrew Schuerger and Rachel Harris
WHAT: They study tough Earth microbes to learn how to spot tiny alien life
WHERE: Venus's clouds, Mars and planets around faraway stars
WHY: Colors and gases from microbes could be clues that life exists

Why it matters

These clues could one day help answer a big question kids often ask: are we alone in the universe?

Test what you remember

1. What kind of alien life do scientists think is most likely?

- (A) Flying people
- (B) Talking animals
- (C) Giant monsters
- (D) Tiny microbes

2. Where on Venus might microbes live?

- (A) On its hot ground
- (B) Under its ice
- (C) High in its clouds
- (D) In its oceans

3. Why do some bacteria make bright colors?

- (A) To look pretty
- (B) To glow at night
- (C) To scare others
- (D) To protect them from strong sunlight

4. What might future telescopes look for on faraway planets?

- (A) Trees
- (B) Colors made by microbes
- (C) Roads
- (D) Cities

5. What gas have rovers found on Mars?

- (A) Helium
- (B) Oxygen only
- (C) Steam
- (D) Methane

6. How deep might a rover need to dig to find Martian microbes?

- (A) About 800 feet
- (B) About 8 miles
- (C) About 8 inches
- (D) It can never dig

Answer key (try the quiz first!): 1: D 2: C 3: D 4: B 5: D 6: C

Questions to think about

1. Andrew Schuerger: Says a rover might only need to dig about 8 inches to find Martian microbes, if they exist.

MY THOUGHTS (at least 20 words):